

TARA, Russian Federation

WMO: 284930

Lat: 56.900N

Lon: 74.383E

Elev: 240

StdP: 14.57

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.3	-27.8	-38.3	0.6	-33.0	-33.0	0.9	-27.6	15.1	16.6	13.7	14.5	1.9	60	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	84.6	67.7	81.6	66.1	78.6	64.7	70.0	80.4	68.2	78.2	66.4	75.8	6.6	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	97.3	76.2	64.4	91.2	73.6	62.6	85.6	71.5	34.2	80.7	32.7	78.1	31.2	75.9	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.4	13.6	11.9	-39.9	89.6	7.3	3.1	-45.1	91.8	-49.4	93.6	-53.5	95.3	-58.8	97.6		
			-39.9	72.8	6.8	2.0	-44.8	74.2	-48.8	75.3	-52.6	76.4	-57.6	77.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.9	-2.2	3.3	19.6	37.0	51.8	62.0	64.9	59.6	49.1	36.8	17.8	4.8
	DBStd	25.80	14.83	13.68	11.05	10.58	9.73	8.42	6.75	6.96	7.66	8.77	14.29	15.24
	HDD50	7274	1619	1307	941	406	98	11	0	5	108	414	965	1400
	HDD65	11546	2084	1727	1406	842	420	151	88	196	479	873	1415	1865
	CDD50	1393	0	0	0	15	154	372	461	304	81	6	0	0
	CDD65	191	0	0	0	0	12	63	84	30	2	0	0	0
	CDH74	1895	0	0	0	5	215	628	743	272	32	0	0	0
	CDH80	435	0	0	0	1	51	153	173	53	4	0	0	0
Wind	WSAvg	5.4	4.9	5.2	6.2	6.4	6.2	5.2	4.6	4.6	5.1	5.7	5.8	5.5
Precipitation	PrecAvg	17.4	0.8	0.5	0.6	1.0	1.5	2.6	2.7	2.4	1.8	1.5	1.1	1.0
	PrecMax	25.2	1.9	1.4	1.3	2.4	4.9	6.9	7.2	5.1	4.3	4.3	2.0	2.1
	PrecMin	11.4	0.2	0.1	0.0	0.1	0.3	0.3	0.2	0.4	0.2	0.4	0.2	0.0
	PrecStd	3.6	0.4	0.3	0.4	0.7	1.0	1.6	1.7	1.1	1.0	0.7	0.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.8	35.2	45.0	72.3	85.4	87.9	88.2	85.5	78.2	63.6	43.8	35.3
		MCWB	30.4	31.8	37.2	55.0	62.4	67.8	70.8	68.3	60.2	50.8	40.1	33.6
	2%	DB	27.9	31.5	40.7	63.7	79.7	84.4	84.3	80.7	71.8	58.0	39.8	32.6
		MCWB	26.5	29.3	34.7	50.5	60.1	66.5	69.2	66.8	58.3	48.4	37.8	31.0
	5%	DB	24.2	28.0	37.6	58.0	75.2	80.8	81.5	76.8	66.6	53.6	36.7	29.0
		MCWB	22.9	26.2	33.1	47.4	58.0	65.6	67.5	64.4	56.0	46.4	34.9	27.5
	10%	DB	19.8	23.6	34.8	53.3	70.3	77.4	78.4	72.9	62.4	49.7	34.1	25.2
		MCWB	18.7	21.7	31.3	44.4	55.8	64.2	66.4	63.3	54.1	44.3	32.6	23.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.7	32.3	38.0	56.2	64.8	72.0	73.0	70.9	63.4	52.7	40.5	33.9
		MCDB	31.7	34.6	43.5	68.8	82.6	83.2	85.4	82.4	74.1	59.0	42.7	34.8
	2%	WB	26.5	29.2	35.4	51.3	61.3	69.0	70.8	68.0	59.5	50.2	37.9	31.1
		MCDB	27.8	31.3	39.2	62.1	76.6	79.3	81.6	77.3	68.8	56.3	39.7	32.4
	5%	WB	23.0	26.2	33.7	48.2	58.8	67.1	69.0	65.9	57.1	47.4	35.0	27.6
		MCDB	24.2	28.0	37.1	57.3	72.0	77.9	79.1	74.4	65.0	52.8	36.3	29.0
	10%	WB	18.7	21.8	31.6	44.9	56.5	65.1	67.2	63.9	54.5	44.5	32.9	23.9
		MCDB	19.8	23.5	34.8	52.6	68.2	75.4	76.4	71.4	61.2	49.6	34.0	25.2
Mean Daily Temperature Range	5% DB	MDBR	15.2	18.5	18.6	17.9	20.9	18.9	17.8	17.6	16.4	12.3	11.4	13.8
		MCDBR	12.6	14.6	17.5	25.5	29.0	24.6	22.6	24.0	24.4	20.3	9.3	12.9
		MCWBR	12.2	13.2	12.9	15.0	13.0	11.5	10.1	11.7	12.9	12.5	8.4	12.6
	5% WB	MCDBR	12.5	14.0	15.2	24.0	26.1	21.5	20.9	20.7	21.9	18.4	9.2	12.7
		MCWBR	12.2	12.9	11.7	14.8	12.8	11.1	10.1	11.3	12.4	12.4	8.6	12.5
Clear-Sky Solar Irradiance	taub	0.278	0.314	0.372	0.412	0.403	0.410	0.427	0.420	0.370	0.352	0.312	0.282	
	taud	2.027	2.011	1.920	2.043	2.235	2.290	2.245	2.238	2.337	2.228	2.072	1.969	
	Ebn at Noon	178	223	238	251	264	264	255	248	245	210	161	135	
	Edh at Noon	19	30	44	45	40	39	40	37	29	25	19	16	
All-Sky Solar Radiation	RadAvg	238	564	1028	1408	1631	1814	1672	1302	869	424	194	143	
	RadStd	24	39	72	128	153	151	121	125	92	47	37	21	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page